Customer Shopping Behaviour Analysis — Project Report

Oct 24, 2026

Executive summary

The analysis of 3,900 customer purchases shows an average spend of $59.76 per purchase, an average review rating of 3.75, and a small subscriber base that is entirely male: no female customer holds a subscription. Clothing is the largest category by both revenue and units sold, followed by Accessories, Footwear and Outerwear. Young adults generate the most purchases and revenue of the four age groups, though the gap between groups is narrow.

The project cleaned the data in Python, stored it in SQLite, answered ten business questions in SQL, and presented the results in a Power BI dashboard. It closes with five recommendations: grow subscriptions, build a loyalty programme, review discount policy, position top products, and target high-value segments.

Objective and scope

The project aims to understand who buys, what they buy, and how they spend, so the business can target marketing, subscriptions and discounts more effectively. It covers demographics (age, gender, location), product mix (item, category, size, colour, season), spending and satisfaction (purchase amount, review rating), and engagement (subscription, discounts, shipping, payment, purchase history and frequency).

Dataset

The dataset `customer_shopping_behavior.csv` holds 3,900 rows and 18 columns, one row per customer purchase, with no duplicate rows.

Group	Columns
Customer	Customer ID, Age (18–70, mean 44.1), Gender (2,652 male, 1,248 female), Location (50 US states)

Group	Columns
Product	Item Purchased (25 items), Category (4), Size, Color (25), Season (4)
Transaction	Purchase Amount (USD), Discount Applied, Promo Code Used, Shipping Type, Payment Method
Behaviour	Review Rating, Subscription Status, Previous Purchases, Frequency of Purchases

The only missing values were 37 blanks in Review Rating (about 0.9% of rows).

Tools and methodology

The work ran as a four-stage pipeline, each stage feeding the next.

1. **Python (pandas, Jupyter)** — loaded, profiled and cleaned the CSV, and engineered new features.
2. **SQLite via SQLAlchemy** — wrote the cleaned data to table `customer_behaviour` in `expense.db`.
3. **SQL (DB Browser for SQLite)** — answered ten business questions with aggregations, subqueries, CASE logic, CTEs and window functions.
4. **Power BI** — built an interactive Customer Behaviour Dashboard with slicers and KPI cards.

Data cleaning and preparation

The raw data needed one imputation, one redundant-column drop, and two new features before analysis.

Step	What was done	Why
Missing values	Filled 37 blank review ratings with the median rating of the same category	Keeps all rows; category median resists outliers
Column names	Lower-cased, spaces replaced with underscores, `purchase_amount_(usd)` renamed to `purchase_amount`	SQL-friendly names

Step	What was done	Why
Age groups	New `age_category` from `pd.qcut` into 4 equal-sized bands: Young_adult, adult, Middle_aged, senior	Enables segment comparison
Purchase frequency	New `purchase_frequency_days` mapping text frequencies to days (Weekly 7, Fortnightly/Bi-Weekly 14, Monthly 30, Quarterly/Every 3 Months 90, Annually 365)	Makes frequency numeric
Redundant column	Dropped `promo_code_used` after confirming it matched `discount_applied` on every row	Removes duplicate information
Export	Loaded the result to SQLite table `customer_behaviour`	Source for SQL and Power BI

SQL analysis

Ten queries on `customer_behaviour` turned the cleaned data into answers to specific business questions.

#	Business question	SQL technique
1	Total revenue from male vs female customers	`SUM` with `GROUP BY gender`
2	Customers who used a discount yet spent at least the average	Scalar subquery on `AVG(purchase_amount)`
3	Top 5 products by average review rating	`AVG`, `ORDER BY`, `LIMIT 5`
4	Average spend: Standard vs Express shipping	`WHERE … IN`, `AVG`
5	Do subscribers spend more? Count, total and average revenue by subscription status	Multiple aggregates in one `GROUP BY`

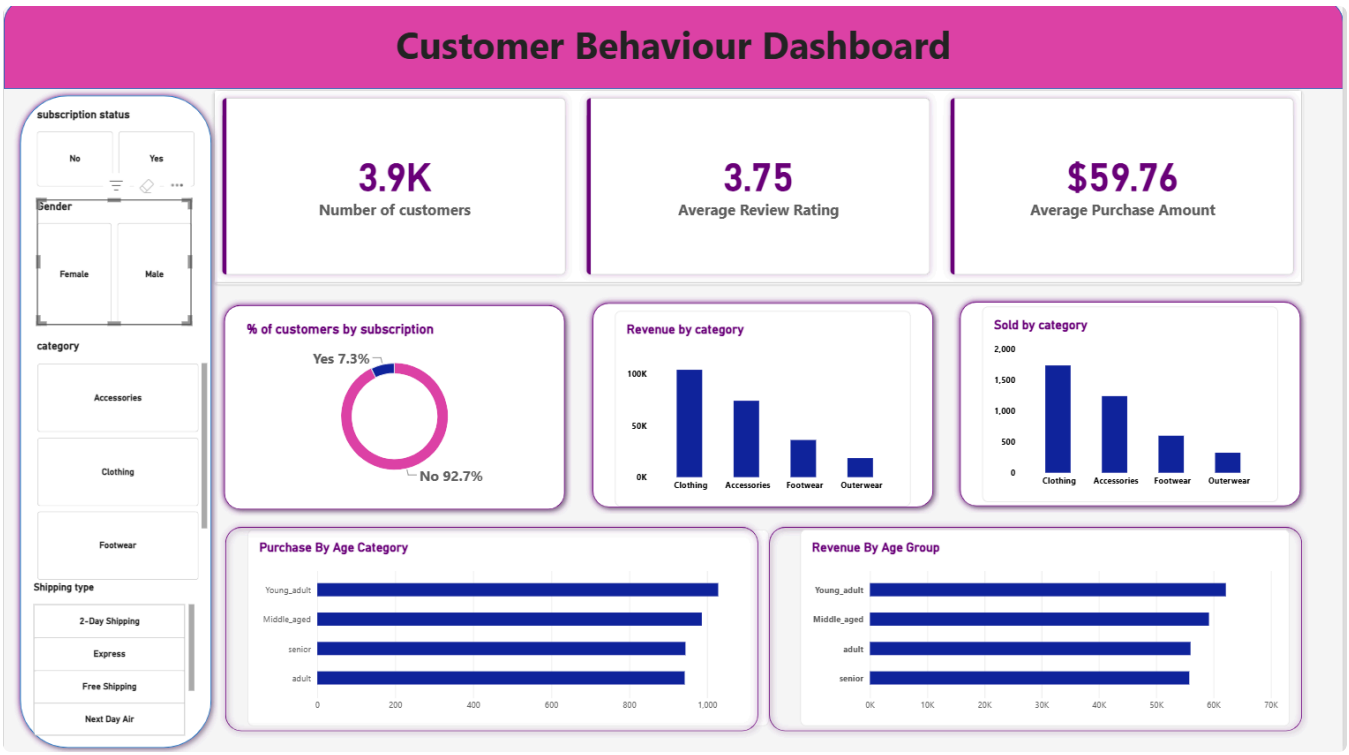
#	Business question	SQL technique
6	Top 5 products by share of purchases with a discount	Conditional count with `CASE WHEN`
7	Customer segments: New (1 previous purchase), Returning (2–10), Loyal (over 10)	CTE with `CASE` segmentation
8	Top 3 most-purchased items within each category	CTE with `ROW_NUMBER() OVER (PARTITION BY category)`
9	Are repeat buyers (over 5 previous purchases) likely to subscribe?	Filtered `COUNT` by subscription status
10	Revenue contribution of each age group	`SUM` by `age_category`, sorted

The full query text is in `expense.sqbpro`. The numeric outputs of each query were not included in the files provided; add them in a result column if the report needs them.

Key finding across queries 1, 5 and 9: every subscribed customer is male, so subscription revenue and repeat-buyer subscription rates come only from men.

Power BI dashboard

The Customer Behaviour Dashboard summarises the data on one page and lets users filter it by subscription status, gender, category and shipping type.



Visual	What it shows
KPI cards	3.9K customers, 3.75 average review rating, $59.76 average purchase amount
% of customers by subscription (donut)	Subscribed vs not subscribed share
Revenue by category	Clothing highest (about $100K), then Accessories, Footwear, Outerwear
Sold by category	Same order by units: Clothing about 1,700, Outerwear lowest at about 320
Purchase by age category	Young_adult leads at just over 1,000 purchases; all four groups are within about 10% of each other
Revenue by age group	Young_adult highest at about $62K; senior lowest at about $55K

Chart values marked "about" are read from the dashboard axes, not exact query outputs.

Key insights

The biggest gap in the business is subscriptions: only 7.3% of customers subscribe, and none of them are women.

1. **No female subscribers.** All subscribed customers are male, although women make up 1,248 of 3,900 customers (32%). The subscription offer is not reaching or appealing to a third of the customer base.
2. **Low overall subscription.** The dashboard shows 92.7% of customers are not subscribed, leaving a large pool to convert.
3. **Clothing drives the business.** It leads in both revenue and units, while Outerwear is the weakest category on both measures.
4. **Age groups are balanced.** Young adults lead purchases and revenue, but the spread between the top and bottom group is under 15%, so no age group can be ignored.
5. **Moderate satisfaction.** An average rating of 3.75 out of 5 leaves room to improve product quality or expectations.
6. **Discounts and promo codes move together.** Every discounted purchase also used a promo code, so the two cannot be evaluated separately.

Business recommendations

The top priority is a subscription offer designed to win female customers, alongside the five broader actions below.

Recommendation	Action	Based on
Boost subscriptions	Promote exclusive subscriber benefits such as early access, free shipping and member pricing	92.7% of customers not subscribed
Win female subscribers	Survey women on why they don't subscribe; test benefits and campaigns aimed at them, starting with Clothing and Accessories	0 of 1,248 female customers subscribed

Recommendation	Action	Based on
Customer loyalty programme	Reward repeat buyers with points or tiers to move them into the Loyal segment (over 10 previous purchases)	SQL query 7 segmentation
Review discount policy	Balance the sales lift from discounts against margin loss; track discounted and full-price spend separately	SQL queries 2 and 6
Product positioning	Feature top-rated and best-selling items in campaigns; review Outerwear range and pricing	SQL queries 3 and 8; category charts
Targeted marketing	Focus spend on high-revenue age groups and Express-shipping customers	SQL queries 4 and 10; age charts

Limitations and conclusion

The findings are solid at the segment level but rest on a single snapshot of 3,900 purchases.

- **No time dimension.** The data has seasons but no dates, so trends over time cannot be measured.
- **One purchase per customer.** Each row is one purchase, so lifetime value relies on the `previous_purchases` count rather than transaction history.
- **Equal-size age bands.** `pd.qcut` splits customers into four equal-count groups, so band boundaries are data-driven rather than standard age ranges.

In conclusion, the business has a healthy, balanced customer base led by Clothing and young adults, but a weak and one-sided subscription programme. Converting even a small share of the 1,248 female customers and the wider non-subscribed base is the clearest growth opportunity.